pre-analysis plan (PAP)

RT (no data group)

Intro remarks:

- We operationalize having daughters vs not having daughters as a comparison between respondents that have one or more daughters vs. respondents that have only one or more sons or no kids at all.
- We opt for an analysis at the individual (not household) level for testing of all hypotheses.

H1: *Having daughters leads to stronger support for gender equality (as measured by attitudes).*That is, denoting *X* as a measure of support for gender equality, (*X* for respondents with daughters) > (*X* for respondents without daughters).

Our constructed variable names (main): “HavingDaughter” (corresponding to construct: Respondents with and without daughters), “GenderAttitude” (corresponding to construct: support for gender equality).

- Variables & their constructions
    - IV hereby named "HavingDaughter" will be derived from the information of each respondent available in *biobirth* which is updated every year. We will use the variables *kidsex01* till *kidsex19*. And code respondents with at least one “2” as a response to these variables as 1 (= having daughter), and respondents without a 2 as a response as 0 (not having a daughter). This will result in one variable denoting whether the respondent has a daughter or not.
    - DV hereby named "GenderAttitude” will be denoted by the item measured in 2018: pl:plh0308_v2: *Best if man and woman work the same amount so they can share the responsibility*. This item is measured on a 7-point Likert scale with 0 = *complete disagree* and 7 = *fully agree*. Any other reported response option will be set to a missing value.
- Data exclusions:
    - We will exclude respondents aged <18 at the time of survey year.
    - We will also exclude respondents who report having their first child in 2018. We do this by excluding “2018” as a response to *biobirth/kidgeb01*.
    - As the variable (item: *pl/plh0308_v2*) used as an indicator of the GenderAttitude is measured in the year 2018 only, we will select the data for analyses of respondents that participated in the panel up until and incl. 2018.
    - Also, we will exclude all those respondents who have a missing value on the GenderAttitude variable (item: *pl/plh0308_v2*)

    - Also, we will exclude all those respondents who will have missing values in any of the control variables (specified below).

- Control variables:
    - BioKids (no. of biological children): biobirth/sumkids. -2 (does not apply) will be recoded as 0.
    - AgeD (age of first daughter) will be determined from biobirth. For those with no daughters, this will be 0.
    - AgeP (age of parent) will be determined from ppathl/gebjahr
    - ParentSex (sex of parent): ppathl/sex
    - NrAdultsH (no. of adults >18 living in a household): We will construct this variable by subtracting the number of kids below 18 (calculated by coding kids in a household, matched through the household ID (hid) based on indicators of children's year of birth such as kidgeb01, kidgeb02 etc, where <18 will be coded as 1 and >18 as 0) from the number of persons living in a household (hgen/hgnrpers)
    - MaritalStatus (marital status) pgen:pgfamstd, all values with a - will be recoded as a missing value
    - PartnerStatus (partner): pgen/pgpartz
    - Edu (respondents education): pgen/pgbilzeit (as 2018 was not measured, we will impute answers from 2017 for that year)
    - Imm (Immigration status): pgen:pgnation (nationality), will we code German as 0 and other as 1
    - EastWest (East/West Germany): ppathl/sampreg (as 2018-2020 was not measured we will impute answers from 2017 for that year)

- Statistical tests:
    - We will use a GLM with a non-normal distribution (reflecting the DV, then applying Gamma distribution with log link) model with HavingDaughter as an independent variable, and GenderAttitude as dependent variable, and control variables as additional fixed effects.
- Interpretation of results:
    - We interpret our result as in line with the hypothesis if the HavingDaughter coefficient is positive, and the *p*-value is <0.05.

H2: *Having daughters leads to more left-leaning political preferences.* That is, denoting *Y* as a measure of left-leaning political preferences, (*Y* for respondents with daughters) > (*Y* for respondents without daughters).

Notes: Here we will use Year (year of measurement of survey) and we will be interested in analysing the causal effect of within-person change, so **whether**

individuals that *gain* a daughter over time show *more left-leaning political preferences.*

- Variables & their constructions:

Each step will be done for each survey year (2005, 2009, 2014, and 2019)

  - IV hereby named "HavingDaughter" will be derived from the information of each respondent available in *biobirth* which is updated every year. We will use the variables *kidsex01* till *kidsex19*. And code respondents with at least one “2” as a response to these variables as 1 (= having daughter), and respondents without a 2 as a response as 0 (not having a daughter).
  - IV named “Left-Right” will be constructed from the variable *pl/plh0004*. The variable is now on a scale from 0-10 and will be reverse-coded such that “very right” = 0 and “very left = 10.
  - IV named “Wave” denoted by the variable *syear.*

- Data exclusions:
  - We will exclude respondents aged <18 at the time of survey year.
  - Since Political Tendency has only been collected in waves 2005, 2009, 2014, and 2019, we will only include responses from these survey years.

- Control variables

Each step will be done for each survey year (2005, 2009, 2014, and 2019)

  - BioKids (no. of biological children): *biobirth/sumkids*. -2 (does not apply) will be recoded as 0.
  - AgeP (age of parent) will be determined from *ppathl/gebjahr*
  - ParentSex (sex of parent): ppathl/sex
  - NrAdultsH (no. of adults >18 living in a household): We will construct this variable by subtracting the number of kids below 18 (calculated by coding kids in a household, matched through the household ID (hid) based on indicators of children's year of birth such as kidgeb01, kidgeb02 etc, where <18 will be coded as 1 and > 18 as 0) from the number of persons living in a household (hgen/hgnrpers)
  - MaritalStatus (marital status) pgen/pgfamstd, all values with a - will be recoded as a missing value
  - PartnerStatus (partner): pgen/pgpartz
  - Edu (respondents education): pgen/pgbilzeit (as 2018 was not measured, we will impute answers from 2017 for that year)
  - Imm (Immigration status): pgen:pgnation (nationality), will we code German as 0 and other as 1

    - EastWest (East/West Germany): ppathl/sampreg (as 2018-2020 was not measured we will impute answers from 2017 for that year)

- Statistical tests:
    - We will use a Linear Fixed Effects Panel Model (FE) with *HavingDaughter* as an independent variable, and *LeftRight* as dependent variable, and control variables as additional fixed effects. We will include **individual and survey year fixed effects** to control for time-invariant individual characteristics and survey wave-specific influences.
- Interpretation of results:
    - We interpret our result as in line with the hypothesis if the HavingDaughter coefficient is positive, and the *p*-value is <0.05. This result will indicate whether having a daughter makes respondents more left-leaning politically (within-person change over time).

H3: *Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).*That is, denoting *Z* as a measure of gender equality in the household, (*Z* for respondents with daughters) > (*Z* for respondents without daughters).

Notes: Here we will use Year (year of measurement of survey) and we will be interested in analysing the causal effect of within-person change, so **whether individuals that *gain* a daughter over time show increased gender equality in the household.**

Our constructed variable names: “HavingDaughter”, “HouseEquality"

- Variables & their constructions

Each step will be done for each survey year (2000-2020)

    - IV hereby named "HavingDaughter" will be derived from the information of each respondent available in biobirth which is updated every year. We will use the variables *kidsex01* till *kidsex19*. And code respondents with at least one “2” as a response to these variables as 1 (= having daughter), and respondents without a 2 as a response as 0 (not having a daughter).
    - IV named “Year” denoted by the variable *syear*.
    - DV hereby named "HouseEquality” will be denoted by a variable indicating the gender equality in housework (1 = full equality, equal share of housework between sexes assuming equal division per

person; 0 = full inequality) in the household of the respondent calculated for every year of the survey (2000-2020). This variable will be calculated with the following steps:

- Step 1: Calculate total housework in a household, TH, sum per person and then sum per household all the reported household work done in Errands (*pl:pli0040*), Hauswork (*pl:pli0043_h*), Childcare (*pl:pli0044_h*), Care for Persons in the week (*pl:pli0046*), Repairs (*pl:pli0049*_h).
- Step 2: Calculate total housework in a household per women (TW) and men (TM), *(ppathl/sex),* sum per person (only women/ only men in the household) and then sum per household all the reported household work done in Errands, Hauswork, Childcare, Repairs, Care for Persons in the week; where T = TW+TM
- Step 3: Calculate Expected fair share: EM(men) = TH*PM(proportion of men in household: PM = Nr of men/ Nr of men+Nr of women), EW(women) = TH*PW (proportion of women in household: PW = Nr of women/ Nr of women+Nr of men)
- Step 4: Calculate final Equality score per household: $1-|Hm-Em|+|Hw-Ew| / T$
- Step 5 - each individual in a household will be matched with its households Equality score

- Data exclusions:

Each step will be done for each survey year (2000-2020)

- We will exclude respondents aged <18 at the time of survey year.
- We will exclude all respondents without a partner in the household.
- Households consisting solely of adults of only one sex will be excluded automatically in the construction of our HouseEquality variable.
- We will exclude respondents with no values on any of the items used to calculate the HouseEquality variable.

- Control variables

Each step will be done for each survey year (2000-2020)

- BioKids (no. of biological children): *biobirth:sumkids*. -2 (does not apply) will be recoded as 0.
- AgeP (age of parent) will be determined from *ppathl/gebjahr*
- ParentSex (sex of parent): ppathl/sex

    - Employment (employment status): actual working hours per week → pgen:l pgtatzeit or pl/plb0186_h; “Does not apply” = -2 should be recoded to 0.
    - NrAdultsH (no. of adults >18 living in a household): We will construct this variable by subtracting the number of kids below 18 (calculated by coding kids in a household, matched through the household ID (hid) based on indicators of children's year of birth such as kidgeb01, kidgeb02 etc, where <18 will be coded as 1 and > 18 as 0) from the number of persons living in a household (hgen/hgnrpers)
    - MaritalStatus (marital status) pgen:pgfamstd, all values with a - will be recoded as a missing value
    - PartnerStatus (partner): pgen/pgpartz
    - Edu (respondents education): pgen/pgbilzeit (as 2018 was not measured, we will impute answers from 2017 for that year)
    - Imm (Immigration status): pgen:pgnation (nationality), will we code German as 0 and other as 1
    - EastWest (East/West Germany): ppathl/sampreg (as 2018-2020 was not measured we will impute answers from 2017 for that year)
    - WorkHours (Actual Weekly Working Hours): pl/plb0186_h. “Does not apply” = -2 should be recoded to 0.

- Statistical tests:
    - We will use a Linear Fixed Effects Panel Model (FE) with HavingDaughter as an independent variable, and HouseEquality as dependent variable, and control variables as additional fixed effects. We will include **individual and survey year fixed effects** to control for time-invariant individual characteristics and survey wave-specific influences.
- Interpretation of results:
    - We interpret our result as in line with the hypothesis if the HavingDaughter coefficient is positive, and the *p*-value is <0.05. This result will indicate whether having a daughter makes the household more equal (within-person change over time).
    - We will use clustered SE with individuals as clusters (pid).

H4: *Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).*That is, denoting *X* as a measure of support for gender equality, (*X* for male respondents with daughters - *X* for male respondents without daughters) > (*X* for female respondents with daughters - *X* for female respondents without daughters).

Our constructed variable names: “HavingDaughter” (corresponding to construct: Respondents with and without daughters), “GenderAttitude” (corresponding to construct: support for gender equality), and “ParentSex”.

- Variables & their constructions
    - IV hereby named "HavingDaughter" will be derived from the information of each respondent available in *biobirth* which is updated every year. We will use the variables *kidsex01* till *kidsex19*. And code respondents with at least one “2” as a response to these variables as 1 (= having daughter), and respondents without a 2 as a response as 0 (not having a daughter). This will result in one variable denoting whether the respondent has a daughter or not.
    - DV hereby named "GenderAttitude” will be measured by the item measured in 2018: *pl/plh0308_v2*: *Best if man and woman work the same amount so they can share the responsibility*. This item is measured on a 7-point Likert scale with 0 = *complete disagree* and 7 = *fully agree*. (any other reported response option will be set to a missing value)
    - We will construct the variable “ParentSex” = using the variable *sex* from the *ppathl* dataset (*ppathl/sex*), recoding the original response options 1 = Male, 2 = Female into 1 = Male, 0 = Female.
- Data exclusions:
    - We will exclude respondents aged <18 at the time of survey year.
    - As the variable (item: plh0308_v2) used as an indicator of the GenderAttitude is measured in the year 2018 only, we will select the data for analyses of respondents that participated in the panel up until and incl. 2018.
    - Also, the analyses will automatically exclude all those respondents who will have missing values in the GenderAttitude variable (item: *pl/plh0308_v2*).
    - We exclude missing values on the variable ParentSex (i.e., all values that are not 1 or 2).
    - Also, we will exclude all those respondents who will have missing values in any of the control variables (specified below).

- Control variables:
    - BioKids (no. of biological children): *biobirth/sumkids*. -2 (does not apply) will be recoded as 0.
    - AgeD (age of first daughter) will be determined from *biobirth*. For those with no daughters, this will be 0.
    - AgeP (age of parent) will be determined from *ppathl/gebjahr*
    - NrAdultsH (no. of adults >18 living in a household): We will construct this variable by subtracting the number of kids below 18 (calculated by

coding kids in a household, matched through the household ID (hid) based on indicators of children's year of birth such as kidgeb01, kidgeb02 etc, where <18 will be coded as 1 and >18 as 0) from the number of persons living in a household (hgen/hgnrpers)
    - MaritalStatus (marital status) pgen:pgfamstd, all values with a - will be recoded as a missing value
    - PartnerStatus (partner) : pgen/pgpartz
    - Edu (respondents education): pgen/pgbilzeit (as 2018 was not measured, we will impute answers from 2017 for that year)
    - Imm (Immigration status): pgen:pgnation (nationality), will we code German as 0 and other as 1
    - EastWest (East/West Germany): ppathl/sampreg (as 2018-2020 was not measured we will impute answers from 2017 for that year)

- Statistical tests:
    - We will use a GLM with a non-normal distribution (reflecting the DV, then applying Gamma distribution with log link) model with HavingDaughter*ParentSex as an independent variable, and GenderAttitude as dependent variable, and control variables as additional fixed effects.
- Interpretation of results:
    - We interpret our result as in line with the hypothesis if the HavingDaughter*ParentSex interaction coefficient is positive, and the *p*-value is <0.05.